

DS203-Programming for Data Science
GROUP PROJECT
EXERCISE -7

Transformer Current Data Cleaning

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DESCRIPTION OF THE PROBLEM

Data Description

The given 'e7-htr-current.csv' file consists of transformer current data from a solar power generation site, sampled at an interval of 5 minutes.

The data points account for about 280 days.

Main Objective

The recorded data consists of missing data as well as noisy and choppy data. The objective in this exercise is to 'fix' the given data and obtain an improved data set with the help of a Machine Learning model created using the 'good' data present in the given data.

Stepwise Tasks

- 01** *Categorizing the days as 'good,' 'bad,' or 'missing' depending on the level of noise in the data for that particular day.*

DESCRIPTION OF THE PROBLEM

02 *Further improving the quality of data on the 'good' days to make them 'really good' by fixing the observed data problems.*

04 *Using the created ML model to 'fix' the data from the 'bad' and the 'missing' days.*

03 *Splitting the good data into 80% training set and 20% testing set; then using the 80% data to create an ML model and the 20% data to validate the model.*

05 *Comparing the "improved data set" with the original one by creating appropriate metrics and analysing their plots.*

SUMMARY OF SOLUTIONS

- *Firstly, we separated 'missing' days and 'bad' days from our data and further cleaned the remaining data to make it ready for analysis.*
- *We fitted different multiple linear regression models on 80% of the data and tested it on 20%, using suitably created new features from the data and analysed the performance of each model.*
- *We selected the model that best captured the variance in the training data while exhibiting minimal overfitting on the test data.*
- *We used the model to predict the current on 'missing' and 'bad' days, thus improved the given data set.*

METHODS EXPLORED FOR CATEGORISATION

- *Visually analysed the data for each day by creating monthly plots on MS excel.*
- *Obtained the sum of absolute difference between consecutive data points for each day using Python and classified the day as 'good', 'bad' or 'missing' by comparing its value with a threshold value.*
- *The threshold was decided manually by us based on the comparison of the classification of data determined visually with those given by the different threshold values.*

IMPROVING THE GOOD DAYS

- *We analyzed the data of good days and found that it contains outliers that do not capture any variance of the data.*
- *So we removed outliers from each day using the method of IQR using a python programme.*
- *After the removal of outliers, 'good' days became 'really good' as they did not contain significant amount of noise anymore.*

CREATING AN ML MODEL: AGGREGATION OF DATA

- *We consolidated the data for each hour of each day by computing the hourly average current. Given the high-frequency nature of the data, using values at every five minutes is unnecessary and can impede model performance. We performed this data aggregation using the functions available in the pandas library.*
- *As per our observations, active hours for each day are between 6am and 6 pm, we created 12 columns for each active hour each having average current values of the hour of a particular day. We used MS EXCEL and pandas functions for it.*
- *After aggregation, the next step was to predict the current at any time of the day by creating an ML model from aggregated data.*

CREATING AN ML MODEL: AGGREGATION OF DATA

- *We needed to predict 12 variables, one for each hour of the day, so we created individual multiple linear regression models for each.*
- *We made line and scatter plots to understand the variation of each hour's current with days, months, week numbers, previous current values etc. using plotly and matplotlib libraries. We made new features from the existing column of date.*
- *We selected different combinations of different independent variables for our model to make ML models for each combination and tested each model using the sklearn library of python.*
- *We compared the models' performances and found the model having x variables as previous current, day number, sq of day no., cube of day no. and fourth power of day no. was the best. So we selected that model.*

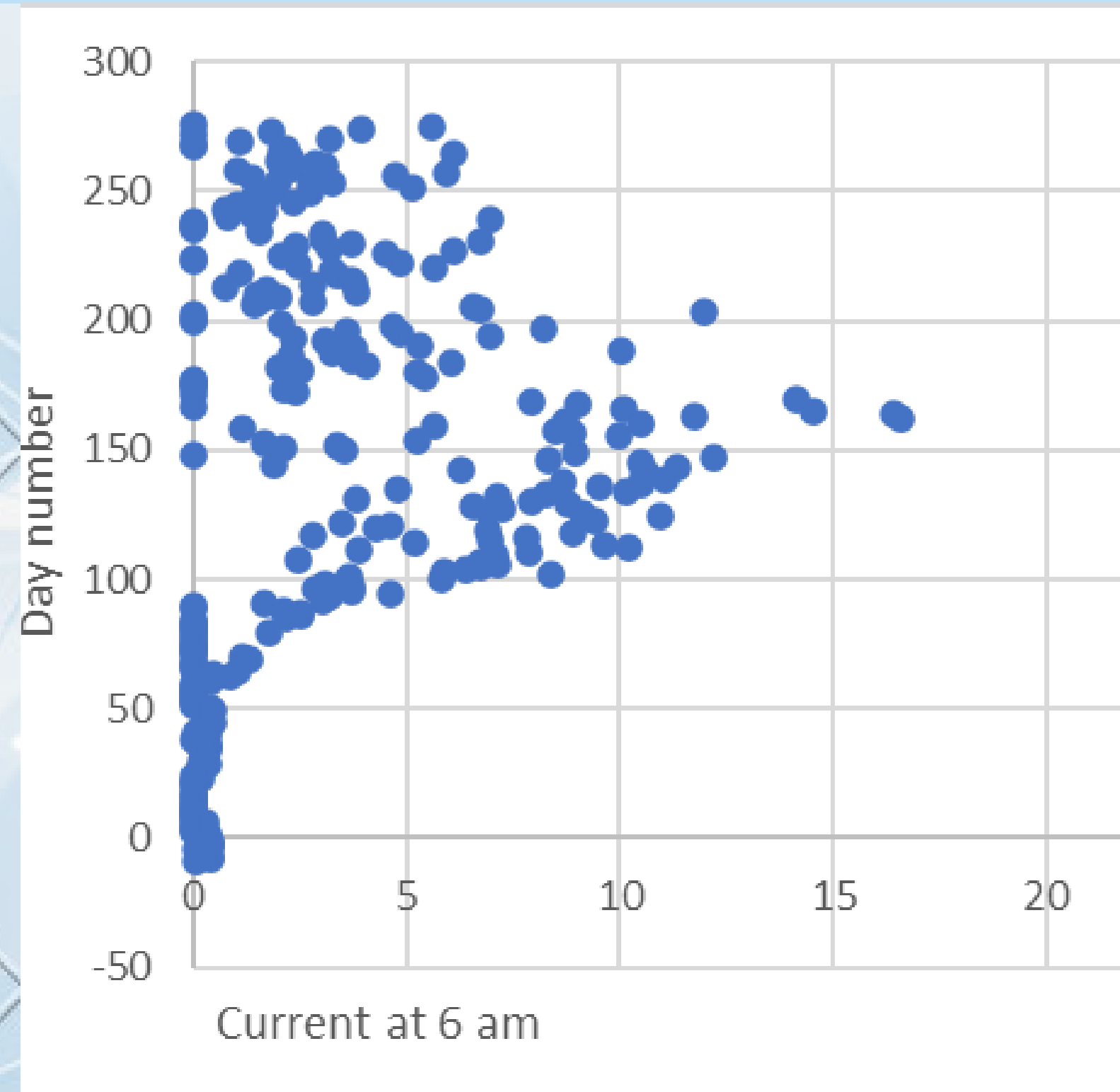
FIXING THE BAD DAYS

- *We first created the same aggregated dataframe for 'bad' and 'missing' days where each row contains the current at different hours of the day.*
- *We filled the data frame with the predicted values of current from our selected ML model.*
- *We then merged the good and bad data frames to get the final data.*
- *In this way the 'bad' days were fixed.*

- *We decided to make ML models of the original data and final data to predict current based on the day number of the year and use the R^2 value of the models as a metric to compare them on the basis of their abilities to be used for an ml model.*
- *We also plotted both the data sets to compare them visually.*
- *The metrics and plots are given in the next slides.*

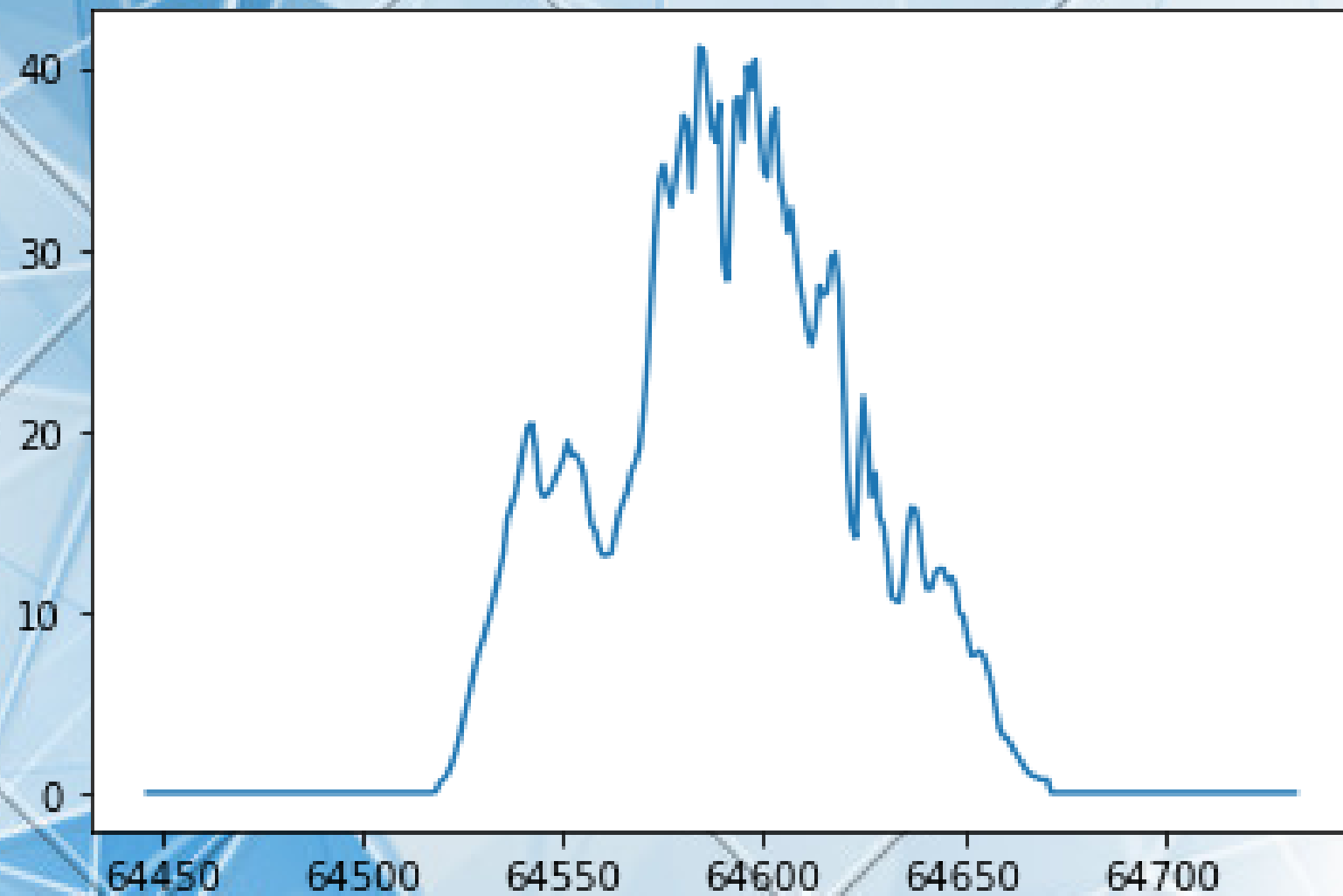
R^2 value	original data	Final data
i at 6	0.64	0.67
i at 7	0.55	0.69
i at 8	0.47	0.79
i at 14	0.41	0.82
i at 15	0.43	0.83

Example of a scatter plot from the original data

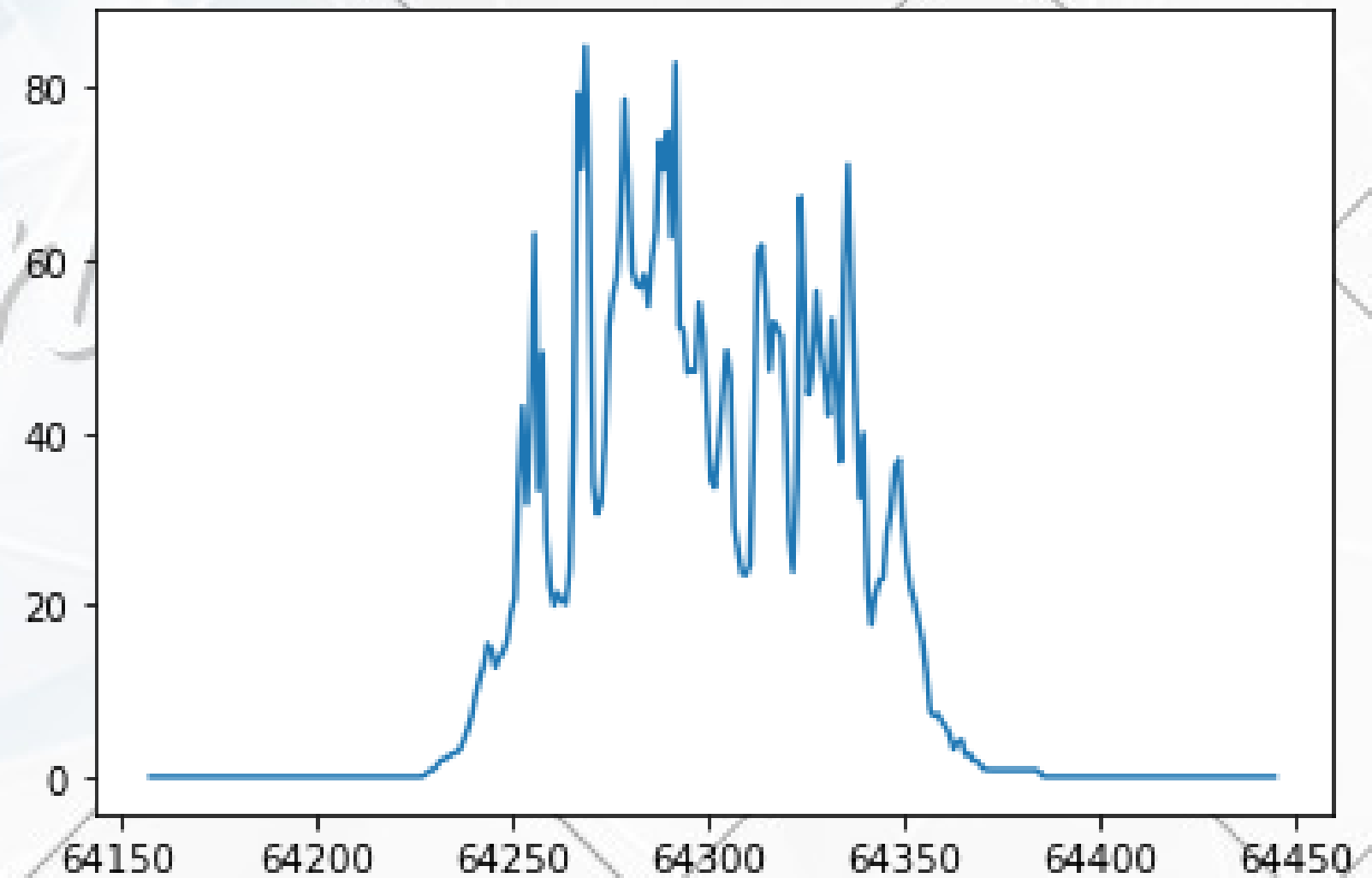


LINE PLOTS

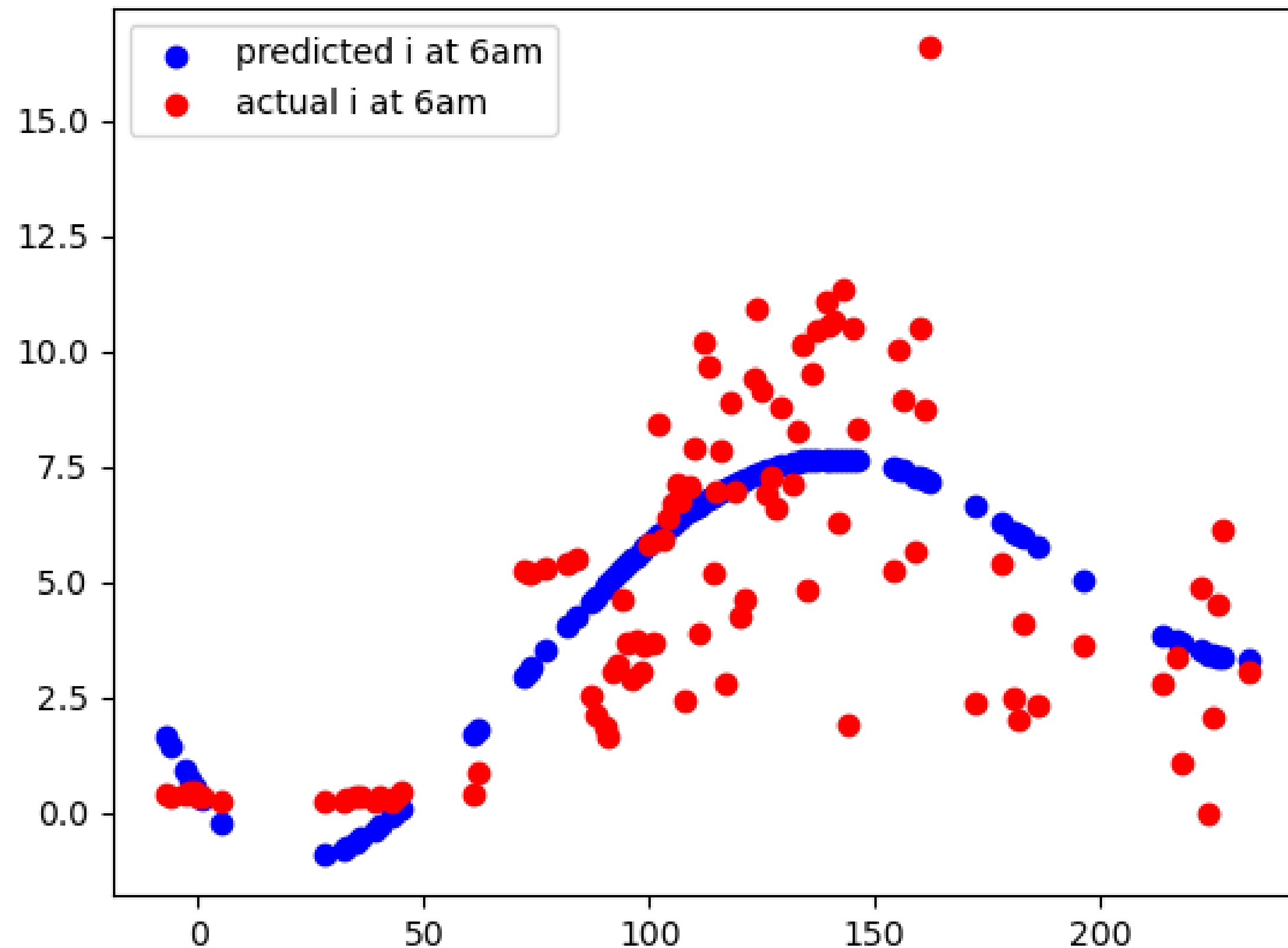
Example of a 'good' day



Example of a 'bad' day



Example of a scatter plot from the final data



OUR APPROACH, WHY IS IT THE BEST?

We tried various approaches like calculating the net deviation of data from moving averages and using the 'sem' function to find the sum of errors from the mean of each day. But the best performance shown for classification was the summing of absolute differences between consecutive data points for each day. The features selected for our ML models also should the best performance compared to the other features.

AN ALTERNATE APPROACH

An appropriate alternate approach would be counting the number of peaks and valleys above a particular threshold for the categorisation of days. Also, we could select some other features for the development of our ML model.

Problems And Their Solutions

Creation of Suitable Features

We created many different features like daynum, week no, day of month, month, etc and finally the features we selected were day no and previous hour's current

Selection of best possible combinations of x variables

The suitable combination of variables were found to be daynum, daynum2, dayno3, dayno4 and prev current

Can such an “improved data set” be used for creating ML models?

Yes, such an "improved data set" is suitable for creating machine learning models.

a. Enhanced Model Performance:

By removing or mitigating noisy or irrelevant data, the improved data set can lead to better model performance. This is because ML models are highly dependent on the quality and relevance of the data they are trained on.

b. Reduced Overfitting:

Noisy data can cause overfitting, where the model fits the training data too closely and doesn't generalize well to new data. A cleaner data set reduces the risk of overfitting.

Thank You